

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION	compare observable properties of soils, rocks and minerals and explain why they are important Earth resources
KEY IDEA	Rocks and minerals can be compared by colour, texture, grain size, hardness and other observable properties. Soil contains weathered rock particles and organic matter; properties influence how resources are used.
WORKED MODEL	Coarse sandy soil drains rapidly because large particles leave bigger spaces; clay-rich soil has tiny particles and usually holds more water.

SUCCESS CRITERIA - I CAN

[] use key vocabulary [] classify or model accurately [] explain using evidence [] correct a misconception

A. BUILD THE SCIENCE IDEA

1. Define and use these terms: soil and rock.

2. Match examples to: soil, rock, mineral.

3. Annotate the worked model with two observations and one scientific explanation.

B. APPLY AND EXPLAIN

4. Sort six examples from this context: sand, clay, garden soil, granite, sandstone and quartz

5. Explain two classifications using observable evidence, not appearance alone.

6. Identify the changed condition, the effect and one condition that should stay the same.

C. INDEPENDENT CHECK

7. Draw or construct a labelled scientific model that shows the relationship.

8. Write a three-sentence explanation using because, therefore and evidence.

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D. APPLY SCIENCE IN CONTEXT

9. Case study: A builder must choose a hard rock for steps while a gardener compares three soils for drainage.

10. Record the important observations in a two-column evidence table.

11. Write a claim-evidence-reasoning conclusion and state one limitation.

E. REASON, EVALUATE AND TRANSFER

12. Find and correct the inaccurate statement in the case study; explain why it is inaccurate.

13. Apply the same science idea to a different everyday situation and predict what would happen.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can compare observable properties of soils, rocks and minerals and explain why they are important Earth resources.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

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Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. Soil is a mixture; rock is solid natural material made of minerals; a mineral has a characteristic composition and properties.
2. Classifications should use observable evidence.
3. Model annotations identify property observations and explain the relationship.
4. Sand/clay/garden soil are soils; granite/sandstone rocks; quartz mineral.
5. Valid reasons include texture, grains, hardness or water behaviour.
6. Changed: sample; measured: drainage/hardness; control: amount, water, time or test method.
- 7-8. Model and explanation must connect particle/property to behaviour.
9. Hard durable rock suits steps; drainage and water-holding evidence guide soil choice.
10. Evidence table records property and implication.
11. Conclusion cites evidence; one small sample is a limitation.
12. Correct any claim that all soils/rocks are identical or that soil is only crushed rock.
13. Prediction must link a property to an appropriate use.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

Soil is not simply dirt or one substance; it is a mixture.

A shiny appearance alone does not prove that a sample is a metal or a particular mineral.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.